Context-Reinforced Random-Field Ensembles for Arbitrary-Resolution Hyperspectral Pansharpening

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***Abstract*—Arbitrary-resolution hyperspectral (ARHS) pansharpening (PNN) uses high-resolution panchromatic (PAN) images to sharpen low spatial resolution HS (LRHS) images to any desired resolutions which may be far beyond the PAN resolution. In contrast, standard hyperspectral (HS) PNN seeks to pansharpen LRHS images to high spatial resolution HS (HRHS) images merely in the PAN resolution. ARHS PNN brings about new challenges for the HS PNN technique, such as PNN scale generalization and PAN-resolution mismatching. In this work, we propose a novel context-reinforced random-field (CR^{2}F) ensemble paradigm for ARHS PNN implicit neural representation (INR), which is able to form strong coordinate-related ensemble feature representation to improve PNN scale generalization and spectral–spatial reconstruction. Then, following the CR^{2}F ensemble paradigm, we further design two instantiated ARHS PNN INRs, i.e., CR^{2}F-INR1 and CR^{2}F-INR2. Experimental results from three datasets verify the excellent performance of our designs.**

***Index Terms*—Arbitrary-resolution hyperspectral (ARHS) pansharpening (PNN), ensemble, hyperspectral (HS) images, random field upsampling, transformer-context.**

NOMENCLATURE

P	Observed PAN image.
H	Observed LRHS image.
$\mathbf{H}^{\uparrow}$	Preinterpolated version of observed LRHS image.
$\mathbf{P}_t$	PAN image for training phase.
$\mathbf{H}_t$	LRHS image for training phase.
$\widetilde{\mathbf{P}}_t$	Synthetic reduced-resolution PAN image for training phase.
$\widetilde{\mathbf{H}}_t$	Synthetic reduced-resolution LRHS image for training phase.
$\widetilde{\mathbf{H}}_t^{\uparrow}$	Preinterpolated version of $\widetilde{\mathbf{H}}_t$.
$\widehat{\mathbf{H}}_a$	Pansharpened HRHS image with an arbitrary resolution.
s	ARHS PNN scale (i.e., spatial resolution ratio between $\widehat{\mathbf{H}}_a$ and $\mathbf{H}$). There is $s = s_d s'$.
s_d	Standard PNN scale (i.e., spatial resolution ratio between $\mathbf{P}$ and $\mathbf{H}$, as the target PNN resolution of standard PNN is the PAN resolution).
s'	Intermediate scale (i.e., spatial resolution ratio between $\widehat{\mathbf{H}}_a$ and $\mathbf{P}$). For standard HS PNN, there are $s' = 1$ and $s = s_d$.

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I. INTRODUCTION

HYPERSPECTRAL (HS) imaging offers great potential for many applications due to its capability to capture spectral and spatial information from the observed scene simultaneously [1], [2], [3], [4]. However, the acquired HS images often have limited spatial resolutions [5], [6]. To improve their spatial resolutions, we usually resort to two ways: one is to design more sophisticated and advanced instruments, and the other is to depend on post-imaging computation. HS pansharpening (PNN) is such an imaging computation technique, which enhances low spatial resolution HS (LRHS) images by integrating the connected panchromatic (PAN) images [7], [8], [9].

The current HS PNN technique can be roughly categorized into two groups: standard HS PNN and arbitrary-resolution HS (ARHS) PNN. The former is to fuse LRHS images and PAN images to produce high spatial resolution HS (HRHS) images with the spatial resolution of PAN images [10], [11]; the latter aims to pansharpen LRHS images to HRHS images with arbitrary spatial resolutions, which could be far beyond the PAN resolution [12], [13].

Almost all existing HS PNN methods are standard ones, which, however, could offer clues for ARHS PNN design. Those standard PNN methods can be approximately attributed to the model-driven group and the deep-learning group. Model-driven methods can further comprise hand-crafted ones and learning-based ones. Component substitution (CS) and multiresolution analysis (MRA) are typical hand-crafted model-driven techniques, which employ fixed mathematical or physical models (such as detail injection, modulation transfer

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function (MTF), and multiscale wavelet/scale spaces) to form some component replacements between LRHS and PAN data [14], [15], [16]; while learning-based model-driven tactic includes variational optimization (VO), Bayesian estimation and matrix factorization [17], [18], [19], [20].

In the past decade, deep-learning-based methods have been widely adopted in the HS PNN task due to their hierarchical learning abilities. Masi et al. [21] innovatively introduced a convolution neural network (CNN)-based structure for multispectral PNN. Then, He et al. [22] proposed HS image specialized CNNs (HyperPNNs), which integrate spectral predictive modules to boost HS PNN. Zheng et al. [23] used a deep prior and dual attention residual network (DARN) for HS PNN. More recently, He et al. [24] proposed dynamic PNN CNNs (DyPNNs), which are able to dynamically learn scene-specific fusion rules. Other representative CNN-based HS PNN include [25], [26], [27], [28], and [29]. Our recent years, transformers are widely used in various tasks such as text generation, image classification and semantic segmentation [30], [31], [32], [33] due to their abilities to capture long-range dependencies, which are also introduced into HS PNN [34], [35], [36]. Bandara and Patel [34] applied the transformer to the context of image PNN first. Then, Zhang et al. [35] proposed the spectral–spatial enhanced transformer-based network (SSETPAN) network for PNN. In addition, Wan and Ma [36] developed a scale-generalized transformer network called DPP, which combines the local feature extraction capabilities of CNNs with the long-range modeling capacity of transformers, enabling more effective PNN across multiple spatial scales.

However, standard HS PNN methods aim to merely enhance LRHS images to the PAN resolution, which obviously restricts their applications where multiple target resolutions or user-customized resolutions are required. Therefore, ARHS PNN has been exploited recently, which brings about new challenges for the HS PNN technique.

Learning-based HS PNN usually works in the training-then-inferring setting. For standard HS PNN, LRHS/PAN, and their self-generated degradation counterparts are used to train a PNN model, and the model is then employed to perform inference from LRHS/PAN so as to produce HRHS data with the PAN resolution under the postulation of PNN scale invariance, i.e., the panshapening scale in the training stage is the same as that in the inferring stage. By contrast, ARHS PNN requires producing HRHS images in any desired resolutions using merely a fixed PNN model, which implies special challenges such as PNN scale generalization and the PAN-resolution mismatching. PNN scale generalization in ARHS PNN means that a trained PNN model under certain PNN scales needs to be generalized to the succeeding inferring stage to produce HRHS images with any PNN scales, which could be far beyond the training scales; while standard HS PNN holds PNN scale invariance across the training and the inferring stages. The PAN-resolution mismatching in ARHS PNN refers to the fact that the target PNN resolution is usually different from the PAN resolution and even far higher than PAN; while in standard HS PNN, the PAN resolution is the same as the target PNN resolution, which implies that the PAN data can be directly injected into LRHS images to enhance them to the PAN resolution. Accordingly, ARHS PNN is obviously more challenging than standard HS PNN. He et al. [12] proposed a CNN-based offset sharing arbitrary-resolution HS PNN framework (ARHS-CNN), which employs a two-stage relay optimization strategy to enable cross-scale reconstruction while preserving spectral fidelity. Building upon the shared spatial patterns across different resolution levels, they further developed a variable subpixel convolution (VSPC-CNN) approach to achieve ARHS PNN [13].

Implicit neural representations (INRs) witness surging interests in tasks of flexibly enhancing data resolutions due to their continuous function considerations [37], [38], [39], [40], [41], [42], [43], [44], which could be utilized for ARHS PNN design since an observed scene is naturally spectral–spatial continuous. Chen et al. [41] proposed a local implicit image function (LIIF) to perform arbitrary-scale image super-resolution. Cao et al. [42] introduced self-attention mechanisms into INR by incorporating both attention maps and spatial distance information to achieve arbitrary-scale image super-resolution. Chen et al. [43] further extended the work with a cascaded attention-based INR.

In this article, our work focuses on designing a new ensemble paradigm for ARHS PNN and accordingly building two specific ARHS PNN INRs. The major consideration in our designs is to strengthen the ensemble features to build strong representations for the coordinate-related data, so as to accomplish better PNN scale generalization and spectral–spatial reconstruction. More specifically, innovative contributions of our work can be identified as follows.

1) A context-reinforced random-field (CR^2F) ensemble paradigm. We develop an innovative ensemble paradigm for ARHS PNN INR, i.e., CR^2F ensemble. Considering the abundant spectral–spatial context information contained in HS/PAN images, CR^2F ensemble builds a two-level context refining process. In the refining process, the spectral–spatial contextual information is first globally extracted on the low-resolution lattice with an elaborated transformer-inspired processing; and then a random field built on the reinforced context features is used to perform an upsampling and therefore to design the final ensemble features. With the two-level paradigm connecting two context extractions, the ensemble is able to sufficiently refine overall contextual information and offer effective ensemble features for the subsequent query mapping.
2) The first CR^2F-oriented ARHS PNN INR (CR^2F-INR1). Following the proposed CR^2F ensemble paradigm, we design the first specific PNN INR instantiation, i.e., CR^2F-INR1, which performs ARHS PNN progressively by decomposing the target PNN scale into a sequence of finer scales. More specifically, in each subscale block, we design a spectral–spatial transformer-based process to extract spectral–spatial context. Building upon that spectral–spatial contextual information, a newly designed random-field upsampling is used to refine the further context information, which forms the final ensemble features using expectation on the random field.

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3) The second CR^{2}F-oriented ARHS PNN INR (CR^{2}F-INR2). We further design the second instantiation of the CR^{2}F ensemble paradigm, i.e., CR^{2}F-INR2. Basically, CR^{2}F-INR2 is also built on sequential finer scales. However, its structure is original input guided and with dual-path random-field upsampling. Therein, the original PAN/LRHS are inserted into each subscale block to strengthen context gathering of a more specially designed transformer process, and then, the spectral context and spatial context are separately fed into their connected random fields to produce final ensemble features. Through the careful processing, more contextual information is expected to be refined in the ensemble step, which is beneficial to the later query mapping.
4) Explicit hierarchical ensemble partition. We stipulate a hierarchical ensemble structure in an explicitly sequential manner, which could guide further ensemble design. The first layer performs context aggregation to capture global or local dependencies, which can be achieved by using a transformer-based architecture, other attention mechanisms, convolutions, or even simpler non-deep learning operations. Based on the preliminarily aggregated information, the second layer focuses on upsampling, which leverages local or global information to predict features at arbitrary query locations, finally acquiring ensemble features.
5) Explicit positional encoding random field and double-explicit positional encoding mechanism. In both CR^{2}F-INR and CR^{2}F-INR2, we explicitly integrate positional information into the covariance parameters of the likelihood of an arbitrary location with respect to low-resolution features, which builds a positional encoding random field. Moreover, we further explicitly encode positional information to improve query mapping. Both low-resolution positional encoding and high-resolution positional encoding are able to strengthen the coupling of positions and the features. The former works in the ensemble process while the latter performs in the succeeding query mapping process. Such a double-explicit positional encoding system designed in CR^{2}F-INR and CR^{2}F-INR2 effectively reinforces the relationship between ensemble features and their corresponding locations, therefore promoting PNN scale generalization and spectral–spatial reconstruction. It is noteworthy that in our positional encoding systems, we use an innovatively designed Gabor-form coordinate function, instead of the traditional Sinusoidal harmonic-based coordinate function.

The rest of our work is divided into several parts. Section II introduces our CR^{2}F paradigm. Sections III and IV give the designs of CR^{2}F-INR1 and CR^{2}F-INR2, respectively. Section V shows our experimental results on several datasets. Section VI is the conclusion of our work.

II. CR^{2}F Paradigm

INRs use neural networks to approximate continuous functions. In the ARHS PNN task, we set the considerations for a practical INR as shown in Fig. 1. An ARHS PNN INR

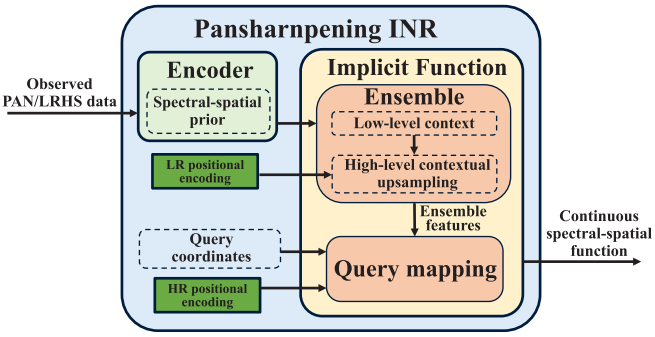


Fig. 1. Overall PNN INR structure.

usually comprises two processes: one is spectral–spatial latent encoding used to concentrate the original input to form a latent code characterizing the prior of the observed scene; the other is an implicit function constructing a prior-based function mapping relative to queried coordinates. More specifically, the implicit function comprises an ensemble procedure and a query mapping. The ensemble processing aggregates useful information from input data so as to constitute ensemble features to boost the final query mapping for arbitrary positions. The ensemble processing usually resides in the core of a PNN INR design.

In the train-then-infer setting, an ARHS PNN INR usually includes a training stage and a succeeding inference stage. The training stage aims to learn the INR model on training data, which can be formulated as

$$\left(\boldsymbol{\theta}_f^{opt}, \boldsymbol{\theta}_\phi^{opt}\right) = \arg\min_{\boldsymbol{\theta}_f, \boldsymbol{\theta}_\phi} \mathcal{L}\left(F\left(\mathbf{z}_t, \mathbf{r}_t = \Phi\left(\widetilde{\mathbf{H}}_t^{\uparrow}, \widetilde{\mathbf{P}}_t; \boldsymbol{\theta}_\phi\right); \boldsymbol{\theta}_f\right), \mathbf{H}_{gt}\right) \tag{1}$$

where $\Phi(\cdot;\boldsymbol{\theta}_\phi)$ is the encoder parameterized by $\boldsymbol{\theta}_\phi$, $F(\cdot;\boldsymbol{\theta}_f)$ is the coordinate-based implicit function network parameterized by $\boldsymbol{\theta}_f$, $\mathcal{L}(\cdot)$ is the loss function, $\mathbf{z}_t = (x_t, y_t, b_t)$ denotes the spectral–spatial coordinates queried on training data, $\mathbf{r}_t$ represents the latent code used as the prior connected to training data, and $\boldsymbol{\theta}_\phi^{opt}$, $\boldsymbol{\theta}_f^{opt}$ represent the optimized parameters obtained on training data. Taking the ensemble process into consideration, (1) can be reexpressed as

$$\left(\boldsymbol{\theta}_e^{opt}, \boldsymbol{\theta}_q^{opt}, \boldsymbol{\theta}_\phi^{opt}\right) = \arg\min_{\boldsymbol{\theta}_e, \boldsymbol{\theta}_q, \boldsymbol{\theta}_\phi} \mathcal{L}\left(F_q\left(\mathbf{z}_t, \mathbf{S} = F_e\left(\boldsymbol{r}_t, \widetilde{\mathbf{H}}_t^{\uparrow}, \widetilde{\mathbf{P}}_t; \boldsymbol{\theta}_e\right); \boldsymbol{\theta}_q\right), \mathbf{H}_{gt}\right) \tag{2}$$

where F_q is the query mapping network parameterized by $\boldsymbol{\theta}_q$, F_e denotes the ensemble processing network, $\boldsymbol{\theta}_e$ is the parameter of the ensemble, and $\mathbf{S}$ stands for the ensemble feature. Accordingly, we have the subsequent inferring stage mathematically represented as

$$\hat{\mathbf{H}}_{\mathbf{a}} = F_q\left(\mathbf{z}_p, \mathbf{S} = F_e\left(\boldsymbol{r}_p, \mathbf{H}^{\uparrow}, \mathbf{P}; \boldsymbol{\theta}_e^{opt}\right); \boldsymbol{\theta}_q^{opt}\right) \tag{3}$$

where $\mathbf{z}_p = (x_p, y_p, b_p)$ denotes the spatial–spectral coordinate queries for inference phase, $\mathbf{r}_p$ stands for the prior of inferring data obtained with the trained parameters $\boldsymbol{\theta}_\phi^{opt}$.

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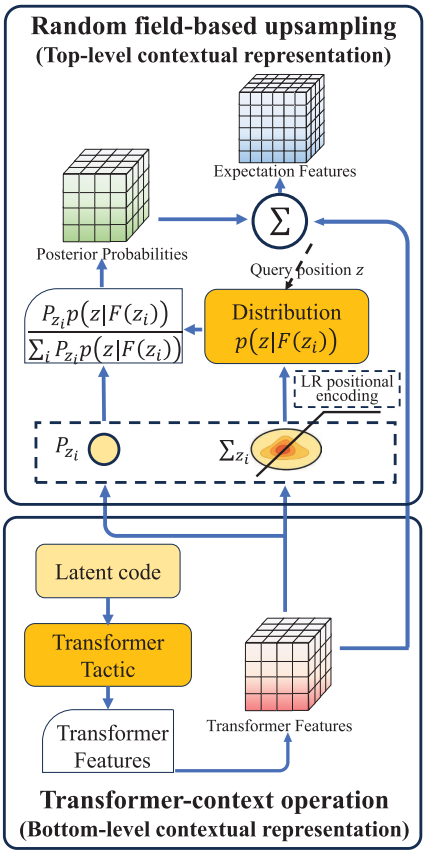


Fig. 2. CR^{2}F ensemble paradigm with hierarchical context representation.

The ensemble design is crucial for ARHS PNN INR. On the one hand, a suitable ensemble processing is able to build more representative ensemble features and enhance the coupling between locations and the ensemble features, thus being expected to improve the PNN scale generalization; on the other hand, the ensemble processing can learn the PAN details shifting across distinct resolutions, thus helping to relieve the PAN-resolution discrepancy.

As shown in Fig. 1, we build a two-level hierarchical design in ensemble by considering to sufficiently mine the abundant spectral–spatial information that PAN/LRHS concentrates into prior. More specifically, we develop a novel CR^{2}F paradigm, whose graphical illustration in more detail can be found in Fig. 2. Spectral–spatial context information reflects the relationship of spectral and spatial dependence inside the data. In our CR^{2}F ensemble paradigm, both of the two levels of context representations are globally built. First, a transformer-based global tactic (GT) is constructed to extract low-level spectral–spatial context as intermediate ensemble features $\mathbf{F}$, which could be formulated as

$$\mathbf{F} = \mathbf{GT}\left(\{\mathbf{Q}_i\}_{i=1,2...}, \{\mathbf{K}_i\}_{i=1,2...}, \{\mathbf{V}_i\}_{i=1,2...}\right) \tag{4}$$

where $\{\mathbf{Q}_i\}_{i=1,2...}$ denotes an array of global queries, $\{\mathbf{K}_i\}_{i=1,2...}$ and $\{\mathbf{V}_i\}_{i=1,2...}$ represent the related global key array and value array, respectively.

Then, in the high-level context representation, we assume that features on various locations are statistically dependent on one another, thus forming a random field point of view. Let $\mathbf{z}$ be an arbitrary-resolution coordinate and $\mathbf{z}'_i$ be the ith low-resolution coordinate. The occurrence of the arbitrary-resolution location $\mathbf{z}$ is assumed to be measured by the following distribution:

$$p\left(\mathbf{z}|\mathbf{F}\left(\mathbf{z}'_i\right)\right) = \frac{1}{2\pi|\Sigma_{\mathbf{F}}|^{1/2}} \exp\left(-\frac{1}{2}\left(\mathbf{z}-\mathbf{z}'_i\right)^T \Sigma_{\mathbf{F}}^{-1}\left(\mathbf{z}-\mathbf{z}'_i\right)\right) \tag{5}$$

where $\mathbf{F}(\mathbf{z}'_i)$ denotes the intermediate features on the coordinate $\mathbf{z}'_i$, and $\Sigma_{\mathbf{F}}$ represents the covariance matrix related to not only the intermediate features but also the low-resolution positions. In such a covariance matrix computation, positions are implicitly encoded into the distribution.

Then, considering that the features on all the low-resolution coordinates contribute to $\mathbf{z}$, we have the posterior occurrence of $\mathbf{F}(z'_i)$

$$P\left(\mathbf{F}\left(\mathbf{z}'_i\right)|\mathbf{z}\right) = \frac{P\left(\mathbf{F}\left(\mathbf{z}'_i\right)\right) p\left(\mathbf{z}|\mathbf{F}\left(\mathbf{z}'_i\right)\right)}{\sum_i P\left(\mathbf{F}\left(\mathbf{z}'_i\right)\right) p\left(\mathbf{z}|\mathbf{F}\left(\mathbf{z}'_i\right)\right)} \tag{6}$$

where

$$P\left(\mathbf{F}\left(\mathbf{z}'_i\right)\right) = \frac{1}{1+\exp\left(-\mathbf{F}\left(\mathbf{z}'_i\right)\right)} \tag{7}$$

is the prior probability of $\mathbf{F}(\mathbf{z}'_i)$, which measures the occurrence or the importance of the features on the ith low-resolution coordinate.

In fact, the posterior probability $P(\mathbf{F}(\mathbf{z}'_i)|\mathbf{z})$ reflects the dependency degree of $\mathbf{z}'_i$'s feature to $\mathbf{z}$'s feature if given the arbitrary coordinate $\mathbf{z}$. Therefore, we can assume that the state space of the arbitrary coordinate's feature $\mathbf{H}(\mathbf{z})$ is $\{\mathbf{F}(\mathbf{z}_i)\}_i$ and the connected probability mass function is posterior probability $P(\mathbf{F}(\mathbf{z}'_i)|\mathbf{z})$. Then, the feature expectation on the arbitrary coordinate $\mathbf{z}$ is of the form

$$\mathbf{H}(\mathbf{z}) = E\left(\mathbf{F}\left(\mathbf{z}'\right)|\mathbf{z}\right) = \sum_i P\left(\mathbf{F}\left(\mathbf{z}'_i\right)|\mathbf{z}\right)\mathbf{F}\left(\mathbf{z}'_i\right) \tag{8}$$

which forms ensemble features for succeeding query mapping.

Our ensemble paradigm adopts a two-level context representation. The bottom-level is connected to a global specialized transformer operation to preliminarily extract global spectral-spatial context information. Formed on the extraction, the top-level builds an upsampling process by computing the random field-based expectation to further statistically strengthen the global context information refining. Such an ensemble intensively refines contextual information from the complex spectral–spatial data. In addition, the positional encoding in random field-based upsampling strengthens the coupling of positions and features. Therefore, our ensemble design is able to effectively build strong coordinate-related ensemble feature representation, which would improve PNN scale generalization and spectral–spatial reconstruction of ARHS PNN.

In the following, we develop two specific ARHS PNN INRs following our CR^{2}F paradigm, i.e., CR^{2}F-INR1 and CR^{2}F-INR2. CR^{2}F-INR1 is a directly intuitive instantiation of the CR^{2}F paradigm, aiming to demonstrate the core idea of the CR^{2}F ensemble paradigm without introducing relatively excessive structures. In contrast, CR^{2}F-INR2 is designed to further enhance PNN performance by explicitly strengthening

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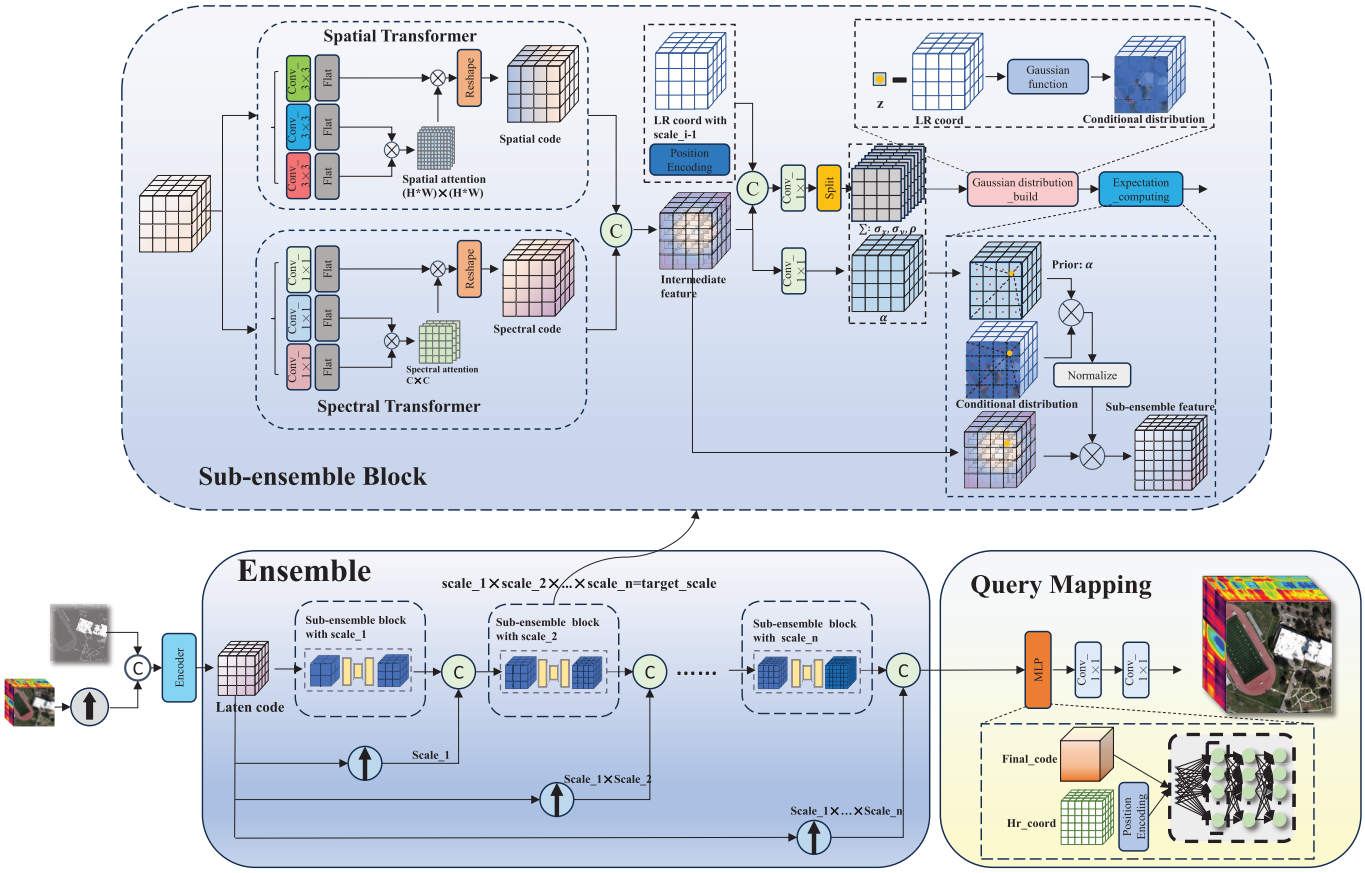


Fig. 3. CR^{2}F-INR1. The lower part of the figure shows the overall architecture of CR^{2}F-INR1. The upper part of the figure shows the architecture of a subensemble block.

the guidance from original observations and separating spatial and spectral context refinement.

III. CR^{2}F-INR1

Fig. 3 graphically illustrates the overall architecture of our CR^{2}F-INR1. It first adopts a convolutional subnetwork as an encoder to extract latent codes. The latent codes are used as the spectral–spatial prior for the succeeding ensemble process and query mapping, where the ensemble design resides in the core of developing CR^{2}F-INR1. More specifically, an ensemble process following our CR^{2}F paradigm is developed. The ensemble process employs a progressive structure, which serializes several subensemble blocks to correspond to decomposing the overall upsampling scale into a sequence of finer scales. Each subensemble block accounts for a fine scale and consists of two successive contextual operations: one is transformer-inspired context extraction, and the other is a random-field processing to contextually perform further upsampling. Skip connections are inserted to the ends of the blocks to retain the original spectral–spatial information (as shown in Fig. 3). A subensemble block can therefore be expressed as

$$\mathbf{S}^k = \text{Fuse}\left[\mathbf{E}^k = \Phi^k\left(\mathbf{S}^{k-1}\right), \mathbf{M}^k\right], \quad k = 1, \ldots, K \tag{9}$$

where k denotes the index of finer scales, $\mathbf{S}^k$ stands for the output feature of subensemble block ($\mathbf{S}^0$ stands for latent code $\mathbf{r}$), $\mathbf{E}^k$ denotes the expectation feature from CR^{2}F-inspired module $\Phi^k(\cdot)$, $\mathbf{M}^k$ is the prior feature by upsampling latent code $\mathbf{r}$ to match the resolution of feature $\mathbf{E}^k$, and Fuse$[\cdot]$ denotes the fusion operation composed of a channel-wise concatenating and a 1×1 convolution.

A. Transformer-Context Operation

Transformer-context operation uses a transformer-inspired tactic to extract low-level contextual information in a subensemble block, which basically consists of a separate spatial transformer and spectral transformer. With the design of separate transformers, spectral–spatial context information can be extracted in relatively low computation burden. The spatial transformer is to refine the global context along the spatial dimension, which is mathematically represented as

$$\begin{aligned} Q'_s &= L_s\left(\mathbf{S}^{k-1}\right), \quad K'_s = L_s\left(\mathbf{S}^{k-1}\right), \quad V'_s = L_s\left(\mathbf{S}^{k-1}\right) \\ Q_s &= FL_s\left(Q'_s\right), \quad K_s = FL_s\left(K'_s\right), \quad V_s = FL_s\left(V'_s\right) \\ A_s &= \text{softmax}\left(\frac{Q_s K_s^{\top}}{\sqrt{C}}\right), \quad \mathbf{F}_s = R\left(V_s A_s\right) \end{aligned} \tag{10}$$

where $L_s(\cdot)$ is the 3×3 convolution, $FL_s(\cdot)$ stands for the flattening operation along the space dimension, A_s denotes the spatial attention, $R(\cdot)$ represents the reshaping operation, and $\mathbf{F}_s$ is the spatial context. In contrast, the spectral transformer

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is used to refine global context along the spectral dimension, which can be formulated as

$$\begin{aligned} &Q_c' = L\left(\mathbf{S}^{k-1}\right), \quad K_c' = L\left(\mathbf{S}^{k-1}\right), \quad V_c' = L\left(\mathbf{S}^{k-1}\right) \\ &Q_c = FL_c\left(Q_c'\right), \quad K_c = FL_c\left(K_c'\right), \quad V_c = FL_c\left(V_c'\right) \\ &A_c = \text{softmax}\left(\frac{Q_c K_c^{\top}}{\sqrt{H \times W}}\right), \ \mathbf{F}_c = R\left(A_c V_c\right) \end{aligned} \tag{11}$$

where $L(\cdot)$ denotes the 1×1 convolution, $FL_c(\cdot)$ is the flattening operation along spectral direction, A_c represents the spectral attention, $R(\cdot)$ denotes the reshaping operation, $\mathbf{F}_c$ is the spectral contextual feature. After the spatial transformer and the spectral transformer processes, the fusion operation comprises channel-wise concatenating and 1×1 convolution

$$\mathbf{F} = \text{Fuse}\left[\mathbf{F}_s, \mathbf{F}_c\right] \tag{12}$$

is used to form the final transformer-context features.

B. Random Field-Based Upsampling

To strengthen the global context refinement in a subensemble block, we design an upsampling operation based on the random field tactic of the CR^{2}F paradigm developed in Section II, which is performed on the obtained transformer-context features.

According to the random field processing in the CR^{2}F paradigm, we first need to calculate the Gaussian conditional distribution of arbitrary query coordinate $\mathbf{z}$ relative to the low-resolution contextual features $\mathbf{F}$ [as mentioned in (5)]. Specifically, the covariance matrix in the conditional distribution is computed based on the context features of the associated low-resolution position, and meanwhile, a position encoding with the low-resolution position is also involved in the computation so as to strengthen the coupling of positions and yielded features (as shown in Fig. 3). For each low-resolution position $\mathbf{z}_i$, we fuse encoded positional information with the contextual feature $\mathbf{F}(\mathbf{z}_i)$, and apply a 1×1 convolution L to generate the Gaussian distribution parameters, which can be formulated as

$$\left[\sigma_{x,i}', \sigma_{y,i}', \rho_i'\right] = L\left(\text{Fuse}\left[\mathbf{F}\left(\mathbf{z}_i\right), \mathcal{P}_l\left(\mathbf{z}_i\right)\right]\right) \tag{13}$$

where $\sigma_{x,i}'$ and $\sigma_{y,i}'$ are raw estimates of the horizontal and vertical variances, ρ_i' represents the associated correlation coefficient, and $\mathcal{P}_l(\cdot)$ stands for the coordinate function used in the positional encoding. Then, the covariance matrix can be represented as

$$\Sigma_i = \begin{bmatrix} \sigma_{x,i}^2 & \rho_i \sigma_{x,i} \sigma_{y,i} \\ \rho_i \sigma_{x,i} \sigma_{y,i} & \sigma_{y,i}^2 \end{bmatrix} \tag{14}$$

where

$$\begin{cases} \sigma_{x,i} = \text{softplus}\left(\sigma_{x,i}'\right) \\ \sigma_{y,i} = \text{softplus}\left(\sigma_{y,i}'\right) \\ \rho_i = \tanh\left(\rho_i'\right) \end{cases} \tag{15}$$

softplus function guarantees $\sigma_{x,i}, \sigma_{y,i} > 0$, and tanh function bounds $\rho_i \in [-1, 1]$ to ensure a proper covariance structure. Summarily, we design a covariance estimation module in the neural network to infer the covariance matrix in an end-to-end manner. The input of the mapping is the combination of low-resolution features and the connected low-resolution positions, which enables the covariance matrix not only to reflect contextual information but also to strengthen the coupling between spatial coordinates and ensemble features. With the inferred covariance matrix Σ_i, we can obtain the Gaussian conditional distribution $p(\mathbf{z}|\mathbf{F}(\mathbf{z}_i))$.

It is worth noting our innovative design of the coordinate function $\mathcal{P}_l(\cdot)$ for the positional encoding. Some of existing literatures insist that Sinusoidal harmonics $\mathcal{P}(x) = [\sin(2^0 x), \cos(2^0 x), \ldots, \sin(2^{l-1} x), \cos(2^{l-1} x)]$ could be used as coordinate function to reduce bias of a neural network to low-frequency components of the processed data [45], [46], [47], [48]. Here, we alternatively develop a novel low-rank Gabor-form [49], [50] coordinate function which employs rank-1 local sin/cos tensor operation to map data onto a more complex manifold, so as to enable the neural network to better fit the high frequency components, which can be formulated as

$$\mathcal{P}_l\left(z\right) = \left\{g_i^{(x)(\theta_j)}\left(x\right) \otimes g_i^{(y)(\theta_j)}\left(y\right)\right\}_{i=0}^{L-1} \tag{16}$$

where

$$g_i^{(x)(\theta_j)}\left(x\right) = \frac{1}{\sqrt{2\pi}\gamma_x} \exp\left(-\frac{x^2}{2\gamma_x^2}\right) \begin{bmatrix} \cos\left(x\omega_{x,i}^{(\theta_j)}\right) \\ \sin\left(x\omega_{x,i}^{(\theta_j)}\right) \end{bmatrix}$$

and

$$g_i^{(y)(\theta_j)}\left(y\right) = \frac{1}{\sqrt{2\pi}\gamma_y} \exp\left(-\frac{y^2}{2\gamma_y^2}\right) \begin{bmatrix} \cos\left(y\omega_{y,i}^{(\theta_j)}\right) \\ \sin\left(y\omega_{y,i}^{(\theta_j)}\right) \end{bmatrix}$$

are rank-1 local sin/cos tensor along x- and y-dimensions, respectively, γ_x and γ_y determine the spatial scales of the Gaussian envelopes of Gabor kernels, $\omega_{x,i}^{(\theta_j)}$ and $\omega_{y,i}^{(\theta_j)}$ are the directional frequencies parameterized by θ_j, $i = 0, \ldots, L-1$ denotes the frequency index, and the operator $\otimes$ denotes the outer product of two tensors [51], [52], [53]. Such an outer product operation on the rank-1 tensors performs separable 2-D filters containing all their pairwise combinations (i.e., $\cos_x \cos_y, \cos_x \sin_y, \sin_x \cos_y, \sin_x \sin_y$).

Based on the above computation, we can further calculate the posterior occurrence $P(\mathbf{F}(\mathbf{z}_i)|\mathbf{z})$ of $\mathbf{F}(\mathbf{z}_i)$ via (6) with the prior $P(\mathbf{F}(\mathbf{z}_i))$ of low-resolution contextual feature, which is formulated as

$$P\left(\mathbf{F}\left(\mathbf{z}_i\right)\right) = \text{Sigmoid}\left(L\left(\mathbf{F}\left(\mathbf{z}_i\right)\right)\right) \tag{17}$$

where $L(\cdot)$ is 1×1 convolution, Sigmoid function guarantees the $P(\mathbf{F}(\mathbf{z}_i))$ as prior validly. Then, the subensemble feature $\mathbf{E}^k$ can be acquired by (8) with the low-resolution contextual feature $\mathbf{F}$ and posterior probability $P(\mathbf{F}(\mathbf{z}_i)|\mathbf{z})$. Therefore, we can further obtain the output feature $\mathbf{S}^k$ via (9). This contextual expectation regression not only achieves smooth and direction-aware feature extraction but also improves the model's ability to generalize across arbitrary resolutions by enhancing the relationship between the global contextual feature and spectral–spatial distance.

After K stages of subensembles, the refined ensemble features $\mathbf{S}^K$ encode spectral–spatial context information. And

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then we need to build a further query mapping to decode ensemble features for restoring the final HRHS image. In the query mapping, we also use the coordinate function designed in (16) to transform the coordinate $\mathbf{z}$ to explicitly enhance the connection between query positions and ensemble features. Then, the query process is further performed with

$$\hat{\mathbf{H}}_a = F_q\left(\mathcal{P}_l\left(\mathbf{z}_{ss}\right), \mathbf{S}^K\right) \tag{18}$$

where $\mathbf{z}_{ss}$ is the set of all high-resolution positions, $F_q(\cdot)$ denotes the spectral–spatial query mapping network composed of a multilayer perceptron (MLP) and two 1×1 convolutions. MLP first produces a preliminary pansharpened result, and then the convolution operations are used to strengthen the reconstruction to generate the final pansharpened result.

Algorithm 1 CR^{2}F-INR1

Input: $\widetilde{\mathbf{H}_t^{\uparrow}}$, $\widetilde{\mathbf{P}}_t$, $\mathbf{H}^{\uparrow}$, $\mathbf{P}$, target scale s
Output: $\widehat{\mathbf{H}}_a$

```
Initialize parameters θ_e, θ_q;
for each epoch do
    Concatenate H̃_t^↑ and P̃_t to form input I;
    Obtain latent code r by encoder network from I;
    Decompose s into K finer scales {s_1, s_2, ..., s_K};
    for i = 1 to K do
        Calculate F^i from F^{i-1};
        Compute Gaussian distribution parameters from F^i;
        Compute E^i following Eq. 5- 8;
        Obtain S^i via Eq. 9 from E^i and r;
    end for
    Obtain S^K with scale s and Ĥ_a by query mapping;
    Compute loss L and update parameters θ_e, θ_q;
end for
Obtain model parameters θ_e^opt, θ_q^opt;
return Ĥ_a = F_q(z; S = F_e(H^↑, P; θ_e^opt), θ_q^opt);
```

The implementation of our CR^{2}F-INR1 is summarized as Algorithm 1.

IV. CR^{2}F-INR2

In addition to CR^{2}F-INR1, we design another CR^{2}F paradigm-based ARHS PNN INR, i.e., CR^{2}F-INR2. Fig. 4 graphically gives its overall architecture. As shown in the figure, the ensemble of our CR^{2}F-INR2 also works in a progressive refining manner. But it is based on dual-path processing and meanwhile uses the input PAN and LRHS images to directly guide context refining. In each subensemble, two branches of processing are built: one is the spatial path, and the other is the spectral path. Each path comprises a cross-transformer-context extraction and a succeeding random field-based upsampling. Such a design is expected to achieve better spectral–spatial context refining. A subensemble block can be formulated as

$$\mathbf{E}^k = \text{Fuse}\left[\Phi_s^k\left(\mathbf{E}^{k-1}, \mathbf{h}^{k-1}\right), \Phi_c^k\left(\mathbf{E}^{k-1}, \mathbf{p}^{k-1}\right)\right],\ k = 1, \ldots, K \tag{19}$$

where $\mathbf{E}^k$ is the output feature of kth subensemble block ($\mathbf{E}_0$ represents latent code extracted by the encoder), $\mathbf{h}^{k-1}$ denotes the upsampled HS image matching the resolution of $\mathbf{E}^{k-1}$, $\mathbf{p}^{k-1}$ denotes the upsampled PAN image, and Φ_s^k and Φ_c^k stand for the CR^{2}F inspired spatial and spectral subensemble branches, respectively.

A. Cross-Transformer-Context Operation

Cross-transformer-context operation seeks to extract low-level contextual information using the transformer tactic across intermediate features and the input LRHS/PAN data. More specifically, in each subensemble, we introduce a latent transformer to extract coarse global context from the input features, and then use further spatial/spectral transformers to combine the coarse context and the LRHS/PAN data.

For the latent transformer, it is used to directly extract global spectral–spatial context information from the spectral–spatial prior. To achieve tractable computation, we design it as a factorized attention module that decomposes the 3-D global model into three 2-D self-attentions along the spectral (B), horizontal (H), and vertical (W) modes [as shown in Fig. (5)].

Taking the horizontal mode (H-mode) for an instance, we apply 1×1 convolutions to generate the query, key, and value in the kth subensemble block

$$\begin{aligned} &Q'_H = L\left(\mathbf{E}^{k-1}\right), \quad K'_H = L\left(\mathbf{E}^{k-1}\right), \quad V'_H = L\left(\mathbf{E}^{k-1}\right) \\ &Q_H = FL_H\left(Q'_H\right), \quad K_H = FL_H\left(K'_H\right), \quad V_H = FL_H\left(V'_H\right) \\ &A_H = \text{softmax}\left(\frac{Q_H K_H^{\top}}{\sqrt{C \times W}}\right),\ \mathbf{f}_H = \text{R}\left(V_H A_H\right) \end{aligned} \tag{20}$$

where $FL_H(\cdot)$ is the flattening operation along the horizontal direction, A_H denotes the horizontal attention map, $R(\cdot)$ is the reshaping operation, and $\mathbf{f}_H$ is the horizontal model global feature.

For the spectral (B) and vertical (W) modes in the latent transformer, we can achieve the spectral and vertical self-attention computing analogously to obtain spectral feature $\mathbf{f}_B$ and vertical feature $\mathbf{f}_W$. The outputs of all three modes are finally fused as the coarse contextual features via

$$\mathbf{I} = \text{Fuse}\left[\mathbf{f}_B, \mathbf{f}_H, \mathbf{f}_W\right]. \tag{21}$$

After the latent transformer, further transformer processing is designed to strengthen the context extraction with the guidance of the LRHS/PAN data. The processing is fulfilled in a two-branch manner. In the spatial transformer path, PAN data and the context features from the latent transformer are integrated to build a PAN-guided spatial context enhancement. In the spectral transformer path, LRHS data and the features from the latent transformer are integrated to build an LRHS-guided spectral context improvement. Given the upsampled PAN image $\mathbf{p}^{k-1}$ and contextual feature $\mathbf{I}$, the cross spatial transformer can be formulated as

$$\begin{aligned} &Q'_s = L_s\left(\mathbf{I}\right), \quad K'_s = L_s\left(\mathbf{p}^{k-1}\right), \quad V'_s = L_s\left(\mathbf{p}^{k-1}\right) \\ &Q_s = FL_s\left(Q'_s\right), \quad K_s = FL_s\left(K'_s\right), \quad V_s = FL_s\left(V'_s\right) \\ &A_s = \text{softmax}\left(\frac{Q_s K_s^{\top}}{\sqrt{C}}\right),\ \mathbf{F}_s = R\left(V_s A_s\right) \end{aligned} \tag{22}$$

where $L_s(\cdot)$ is the 3×3 convolution, $FL_s(\cdot)$ is the flattening operation along the space dimension, A_s is the spatial attention,

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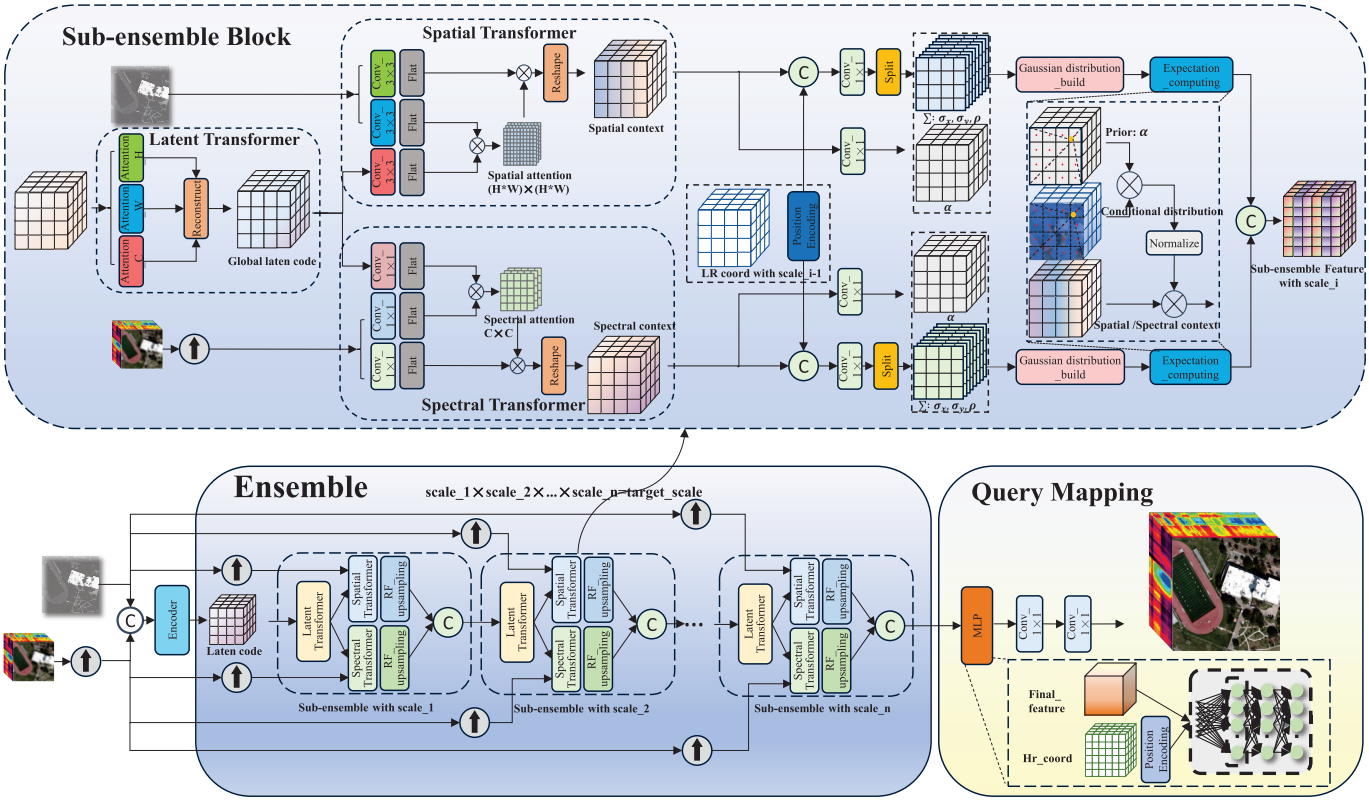


Fig. 4. CR^{2}F-INR2. The lower part of the figure shows the overall architecture of CR^{2}F-INR2. The upper part of the figure shows the architecture of a subensemble block.

and $R(\cdot)$ represents the reshaping operation, which is used to obtain the spatial contextual feature $\mathbf{F}_s$. In contrast, the cross-spectral transformer is designed to refine spectral dependencies by leveraging the upsampled HS image $\mathbf{h}^{k-1}$ and coarse context $\mathbf{I}$, which can be expressed as

$$\begin{aligned} &Q_c' = L(\mathbf{I}), \quad K_c' = L\left(\mathbf{h}^{k-1}\right), \quad V_c' = L\left(\mathbf{h}^{k-1}\right) \\ &Q_c = FL_c\left(Q_c'\right), \quad K_c = FL_c\left(K_c'\right), \quad V_c = FL_c\left(V_c'\right) \\ &A_c = \text{softmax}\left(\frac{Q_c K_c^{\top}}{\sqrt{H \times W}}\right), \ \mathbf{F}_c = R(V_c A_c) \end{aligned} \tag{23}$$

where $L(\cdot)$ denotes 1×1 convolution, $FL_c(\cdot)$ is the flattening operation along spectral direction, A_c is the spectral attention, and $\mathbf{F}_c$ stands for the cross spectral contextual feature.

B. Dual Random Field-Based Upsampling

The ensemble of our CR^{2}F-INR2 is basically a two-path structure. Therefore, the spatial context features and the spectral context features output from the cross-transformer-context operation are separately fed into two random field processes, which establish the spatial branch and the spectral branch of a subensemble. Then, we could perform operations in both spectral and spatial branches to accomplish random field-based upsampling according to our CR^{2}F paradigm.

More specifically, the raw variance parameters of conditional distributions comprising positional encodings for the spatial branch and the spectral branch can be mathematically represented as

$$\left[\sigma_{x,i}^{(spa)'}, \sigma_{y,i}^{(spa)'}, \rho_i^{(spa)'}\right] = L\left(\text{Fuse}\left[\mathbf{F}_s(\mathbf{z}_i), \mathcal{P}_l(\mathbf{z}_i)\right]\right) \tag{24}$$

and

$$\left[\sigma_{x,i}^{(spe)'}, \sigma_{y,i}^{(spe)'}, \rho_i^{(spe)'}\right] = L(\text{Fuse}\left[\mathbf{F}_c(\mathbf{z}_i), \mathcal{P}_l(\mathbf{z}_i)\right]) \tag{25}$$

respectively. Then, we can obtain the spatial covariance matrix

$$\Sigma_i^{(spa)} = \begin{bmatrix} \sigma_{x,i}^{(spa)2} & \rho_i^{(spa)}\sigma_{x,i}^{(spa)}\sigma_{y,i}^{(spa)} \\ \rho_i^{(spa)}\sigma_{x,i}^{(spa)}\sigma_{y,i}^{(spa)} & \sigma_{y,i}^{(spa)2} \end{bmatrix} \tag{26}$$

where

$$\begin{cases} \sigma_{x,i}^{(spa)} = \text{softplus}\left(\sigma_{x,i}^{(spa)'}\right) \\ \sigma_{y,i}^{(spa)} = \text{softplus}\left(\sigma_{y,i}^{(spa)'}\right) \\ \rho_i^{(spa)} = \tanh\left(\rho_i^{(spa)'}\right) \end{cases}. \tag{27}$$

softplus and tanh functions are utilized to maintain the region of these parameters. The spectral covariance matrix $\Sigma_i^{(spe)}$ can be computed in a similar way. With the spatial covariance matrix $\Sigma_i^{(spa)}$ and spatial context $\mathbf{F}_s$, we can obtain the spatial conditional distribution $P_s(\mathbf{z}|\mathbf{F}(\mathbf{z}_i))$ via (5). Then, the prior of low-resolution $P_s(\mathbf{F}(\mathbf{z}_i))$ can be formulated as

$$P_s\left(\mathbf{F}(\mathbf{z}_i)\right) = \text{Sigmoid}\left(L\left(\mathbf{F}_s(\mathbf{z}_i)\right)\right) \tag{28}$$

which is then combined with the spatial conditional distribution $P_s(\mathbf{z}|\mathbf{F}(\mathbf{z}_i))$ to acquire the posterior probability $P_s(\mathbf{F}(\mathbf{z}_i)|\mathbf{z})$

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TABLE I
PROPERTIES OF DATASETS USED FOR EVALUATION

Datasets	Spectral Range	Original HS/PAN	Training HS/PAN	Inferring HS/PAN	Instrument/Source
Changping	0.4-2.5 μm	2900×180×137/8700×540	1200×180×137/3600×540	60×60×137/180×180	Hyperion and ALI
Merced	0.4-2.5 μm	2100×180×134/6300×540	600×150×134/1800×450	60×60×134/180×180	Hyperion and ALI
Pavia Center	0.4-0.8 μm	1000×625×102/	250×100×102/1000×400	50×50×102/200×200	ROSIS

via (6). Therefore, we can achieve the spatial random-field upsampling and acquire the spatial expectation $\mathbf{E}_s$ according to (8). In addition, the spectral random-field upsampling is with the similar processing of the spatial one, which is conducted on the spectral context $\mathbf{F}_c$ and spectral covariance matrix $\Sigma_i^{(spe)}$ to obtain the final spectral expectation $\mathbf{E}_c$. Then, the subensemble features $\mathbf{E}^k$ can be obtained by

$$\mathbf{E}^k = \text{Fuse}\left([\mathbf{E}_s, \mathbf{E}_c]\right). \tag{29}$$

After the K stage subensemble, we can obtain the final ensemble feature $\mathbf{E}^K$ with target PNN scale, then the query mapping can be represented as

$$\hat{\mathbf{H}}_a = F_q\left(\mathcal{P}_l\left(\mathbf{z}_{ss}\right), \mathbf{E}^K\right)$$

where $\mathcal{P}_l(\cdot)$ is the Gabor-form coordinate function, $\mathbf{z}_{ss}$ is the set of all high-resolution positions, $F_q(\cdot)$ denotes the spectral–spatial query mapping subnetwork composed of MLP and two 1×1 convolution operations.

Algorithm 2 CR^{2}F-INR2

Input: $\widetilde{\mathbf{H}_t^{\uparrow}}$, $\widetilde{\mathbf{P}}_t$, $\mathbf{H}^{\uparrow}$, $\mathbf{P}$, target scale $\boldsymbol{s}$
Output: $\widehat{\mathbf{H}}_a$

Initialize parameters $\boldsymbol{\theta}_{\text{e}}$, $\boldsymbol{\theta}_{\text{q}}$;
for each epoch **do**
  Concatenate $\widetilde{\mathbf{H}_t^{\uparrow}}$ and $\widetilde{\mathbf{P}}_t$ to form input $\mathbf{I}$;
  Obtain latent code $\mathbf{r}$ by encoder network from $\mathbf{I}$;
  Decompose s into K finer scales $\{s_1, s_2, \ldots, s_K\}$;
  for $i = 1$ to K **do**
    Calculate $\mathbf{I}^i$ from $\mathbf{E}^{i-1}$;
    Calculate $\mathbf{F}_c^i$ and $\mathbf{F}_s^i$ from $\mathbf{I}^i$ and $\widetilde{\mathbf{H}_t^{\uparrow}}$, $\widetilde{\mathbf{P}}_t$;
    Compute Gaussian distribution parameters based on $\mathbf{F}_c^i$ and $\mathbf{F}_s^i$;
    Compute $\mathbf{E}_s^i$ and $\mathbf{E}_c^i$ following Eq. 5- 8;
    Obtain $\mathbf{E}^i$;
  end for
  Obtain $\mathbf{E}^K$ with scale s and $\widehat{\mathbf{H}}_a$ by query mapping;
  Compute loss $\mathcal{L}$ and update parameters $\boldsymbol{\theta}_{\text{e}}$, $\boldsymbol{\theta}_{\text{q}}$;
end for
Obtain model parameters $\boldsymbol{\theta}_{\text{e}}^{opt}$, $\boldsymbol{\theta}_{\text{q}}^{opt}$;
return $\widehat{\mathbf{H}}_a = F_q(z; \boldsymbol{S} = F_e(\mathbf{H}^{\uparrow}, \mathbf{P}; \boldsymbol{\theta}_{\text{e}}^{opt}), \boldsymbol{\theta}_{\text{q}}^{opt})$;

The implementation of CR^{2}F-INR2 is summarized as Algorithm 2.

V. EXPERIMENTAL RESULTS

In this section, experiments are conducted to assess the effectiveness of the proposed CR^{2}F-INR1 and CR^{2}F-INR2.

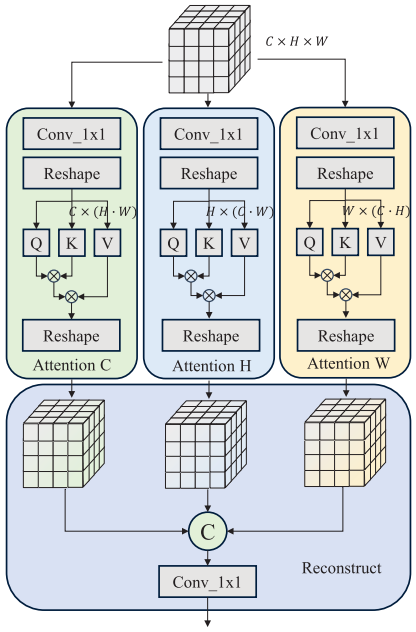


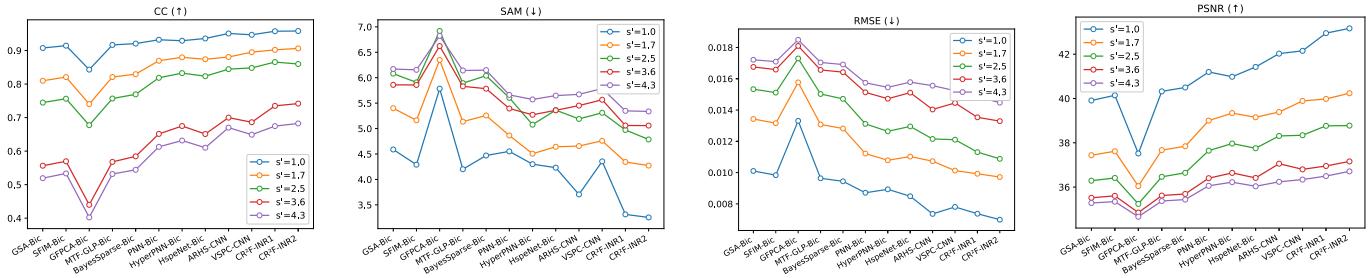
Fig. 5. Structure of latent transformer.

A. Test Datasets and Experimental Setting

We test our proposed ARHS PNN methods CR^{2}F-INR1 and CR^{2}F-INR2 on three HS datasets, including the Changping dataset, the Merced dataset, and the Pavia Center dataset. It is noteworthy that the ARHS PNN scale is the product of the standard PNN scale and intermediate scale, i.e., $s = s_d s'$. The Changping dataset and the Merced dataset are used as real datasets that include HS images and corresponding PAN images, wherein the ratios of PAN to LRHS resolutions are 3, i.e., there is $s_d = 3$ for Changping and Merced. While the Pavia Center dataset is used as a simulated dataset without real PAN images, where PAN images are produced from HS data and the ratio of PAN to LRHS is set to 4 in our experiments, i.e., there is $s_d = 4$ for the Pavia Center dataset. Details of the three test datasets are illustrated in Table I. In addition, the training regions of those datasets are specifically divided into patches in the size of 20 × 20.

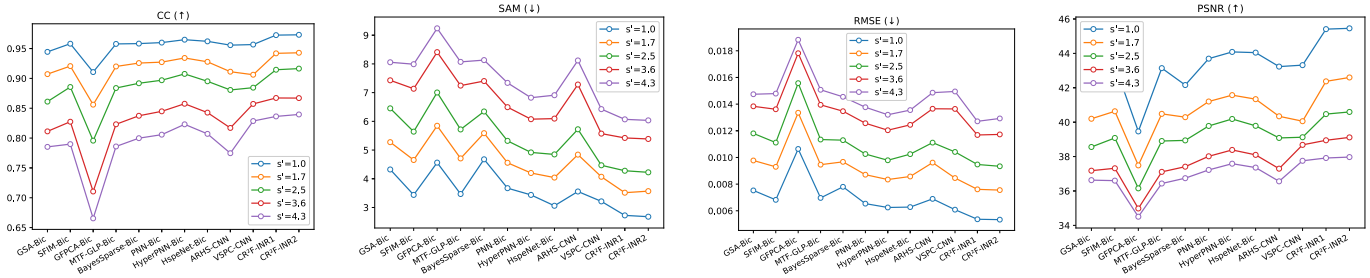
To test the effectiveness of our methods, ARHS PNN experiments consist of reduced-resolution assessment [10], [54], [55], and full-resolution assessment [56], [57]. In the reduced-resolution assessment, quality metrics are evaluated by comparing the observed LRHS image with the PNN result obtained from the synthetic reduced-resolution LRHS and the corresponding PAN image. Four commonly used quality metrics for reduced-resolution assessment are employed,

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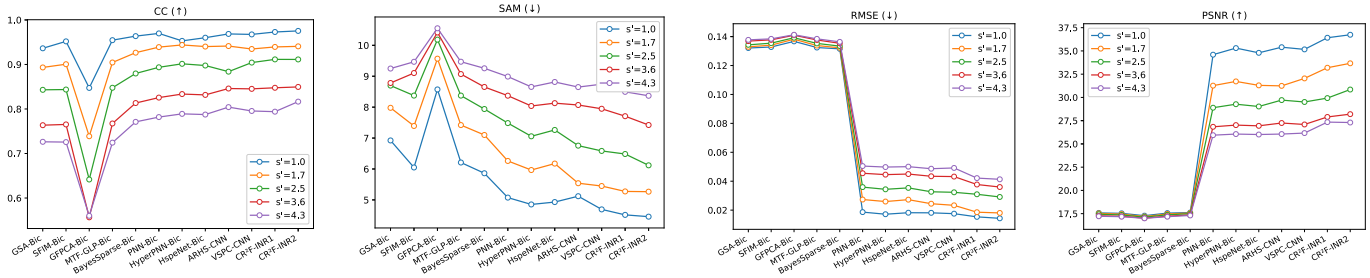
(a) (b) (c) (d)

Fig. 6. PNN performances given an arbitrary scale on the reduced-resolution Changping. (a) CC. (b) SAM. (c) RMSE. (d) PSNR.



(a) (b) (c) (d)

Fig. 7. PNN performances given arbitrary scales on the reduced-resolution Merced. (a) CC. (b) SAM. (c) RMSE. (d) PSNR.



(a) (b) (c) (d)

Fig. 8. PNN performances given arbitrary scales on the reduced-resolution PaviaC. (a) CC. (b) SAM. (c) RMSE. (d) PSNR.

including cross correlation (CC), peak signal-to-noise ratio (PSNR), root-mean-square error (RMSE), and spectral angle mapper (SAM). While full resolution assessment considers the ARHS PNN evaluation directly by the PNN result from the observed HS and PAN image. Specifically, the quality metrics for full-resolution assessment include spatial distortion index D_s, spectral distortion index D_λ, and the quality with on reference (QNR).

Ten existing HS PNN methods are used to compare with our proposed CR^{2}F-INR1 and CR^{2}F-INR2. Five of these methods are model-driven methods, including GSA, SFIM, MTF-GLP, BayesSparse, and GFPCA [8], [14], [19], [58], [59], and three of them are CNN-based methods, including PNN [21], HyperPNN [22], and HSpeNet [60]. While the above eight methods are designed for the standard PNN task, they cannot be directly used in our ARHS PNN experiments. Therefore, to enable these methods to perform ARHS PNN, a bicubic interpolation operation needs to be added to these models as a final PNN step, which is called X-Bic, like GSA-Bic. In addition, we also consider two ARHS PNN methods in our ARHS PNN experiments, concluding ARHS-CNN [12] and VSPC-CNN [13]. Specifically, in VSPC-CNN and our proposed methods, DARN [23] is used as an encoder to extract spectral–spatial information preliminarily. Therefore, a total of 12 methods are used to be compared in our ARHS PNN experiments.

We conducted all the experiments on an NVIDIA V100 Tensor Core GPU, with the Intel Xeon Processor (Skylake, IBRS). The realization of those traditional methods depends on the codes of publication, and deep learning methods, including PNN, HyperPNN, HSpeNet, ARHSCNN, and VSPC-CNN, are conducted strictly following the description of their publications. Furthermore, for all the deep-learning methods, the iterations of the training stage are set to be 300, and the optimizer is AdamW with the initial learning rate 1×10^{-3} [61], [62]. Specifically, the activation function of all the

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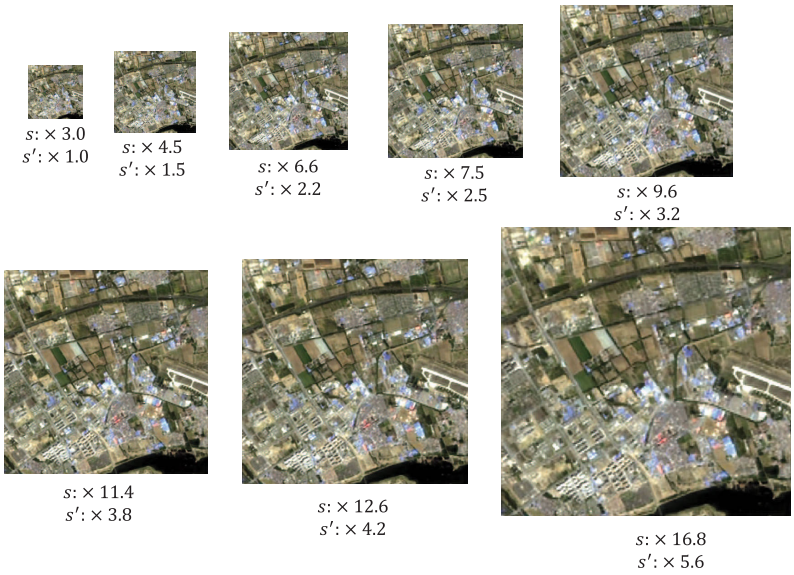


Fig. 9. Visualization of ARHS PNN by CR^{2}F-INR2 on the reduced-resolution Changping, where excellent visual effect is confirmed.

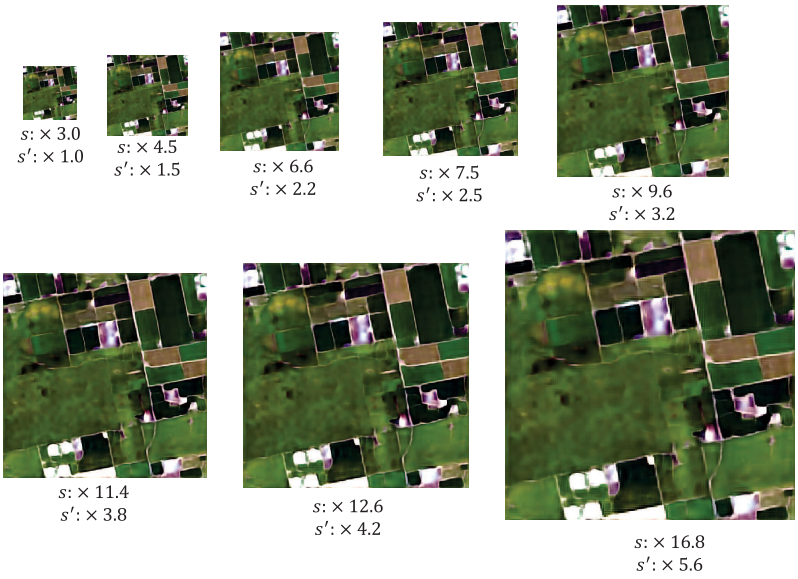


Fig. 10. Visualization of ARHS PNN by CR^{2}F-INR2 on the reduced-resolution Merced, where excellent visual effect is confirmed.

deep-learning methods is the RELU function, and the loss function used in the experiments is the l_1-norm function.

B. Reduced-Resolution ARHS PNN Experiments

We first conduct the ARHS PNN experiments at the reduced-resolution level. In the experiments, reduced-resolution images degraded from observed LRHS images and PAN images are used as input data, and the observed LRHS images are used as ground-truth images. Specifically, in the training stage, the reduced-resolution images are generated over a range of intermediate scales s' from 1.0 to 2.5 with a step of 0.1, and are employed to learn a PNN model. While in the inferring phase, we set $s' = 1.0, 1.7, 2.5, 3.6, 4.3$ and $s' = 1.0, 1.5, 2.2, 2.5, 3.2, 3.8, 4.2, 5.6$ as intermediate scales to produce the inferring data, and test the trained PNN model on the data to make quantitative assessment comparison and visual evaluation, respectively. In Changping and Merced, the ratios of PAN to LRHS resolution are 3, and thus there is the final PNN scale $s = 3s'$ for the two datasets. In Pavia Center, the resolution ratio of PAN to LRHS is set to 4, and therefore we have $s = 4s'$ for this dataset. It is noteworthy that when $s' = 1.0$ (the target PNN resolution is the same as the PAN resolution) and training scales comprise the inferring scale, the ARHS PNN is degenerated to the standard PNN task. That is, our ARHS PNN experiments in fact include standard PNN experiments ($s' = 1.0$).

With ground-truth images being available, ARHS PNN results can be assessed by quantitative quality metrics. Figs. 6–8 display the quantitative results on the Changping dataset, the Merced dataset, and the Pavia Center dataset. Obviously, our CR^{2}F-INR1 and CR^{2}F-INR2 outperform the other methods on the CC, SAM, RMSE, and PSNR

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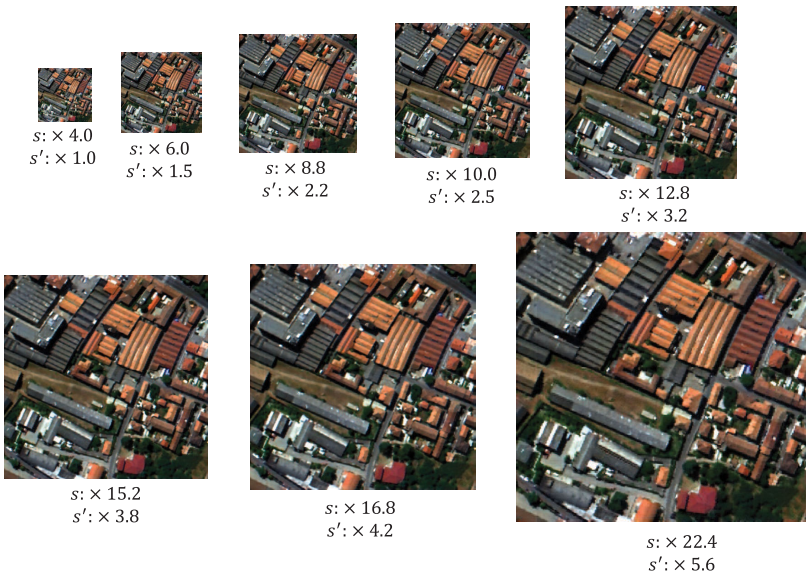


Fig. 11. Visualization of ARHS PNN by CR^{2}F-INR2 on the reduced-resolution PaviaC, where excellent visual effect is confirmed.

TABLE II
QUALITY METRICS OF FULL-RESOLUTION EXPERIMENT ON THE CHANGPING DATASET

Methods	$D_\lambda(\downarrow)$	$D_s(\downarrow)$	QNR(↑)
GSA-Bic	0.0409	0.1530	0.8124
SFIM-Bic	0.0702	0.1067	0.8870
MTF-GLP-Bic	0.0636	0.1408	0.8046
GFPCA-Bic	0.0428	0.2151	0.7514
BayesSparse-Bic	0.0275	0.0761	0.8985
PNN-Bic	0.0750	0.0293	0.8979
HyperPNN-Bic	0.0725	0.0382	0.8921
HSpeNet-Bic	0.0460	0.0345	0.9211
ARHS-CNN	0.0520	0.0175	0.9315
VSP-CNN	0.0669	0.0327	0.9026
CR^{2}F-INR1	**0.0412**	**0.0108**	**0.9484**
CR^{2}F-INR2	**0.0333**	**0.0091**	**0.9579**

TABLE III
QUALITY METRICS OF FULL-RESOLUTION EXPERIMENT ON THE MERCED DATASET

Methods	$D_\lambda(\downarrow)$	$D_s(\downarrow)$	QNR(↑)
GSA-Bic	0.0312	0.0574	0.9132
SFIM-Bic	0.0363	0.0292	0.9356
MTF-GLP-Bic	0.0246	0.0384	0.9379
GFPCA-Bic	0.0394	0.0380	0.9240
BayesSparse-Bic	0.0573	0.0687	0.8779
PNN-Bic	0.0889	0.0167	0.8959
HyperPNN-Bic	0.0550	0.0217	0.9246
HSpeNet-Bic	0.0240	0.0219	0.9546
ARHS-CNN	0.0200	0.0219	0.9583
VSP-CNN	0.0508	0.0145	0.9355
CR^{2}F-INR1	**0.0184**	**0.0132**	**0.9686**
CR^{2}F-INR2	**0.0191**	**0.0094**	**0.9717**

metrics for all five PNN scales. In addition, the bicubic-added models, including GSA-Bic, SFIM-Bic, MTF-GLP-Bic, BayesSparse-Bic, GFPCA-Bic, PNN-Bic, HyperPNN-Bic, and HSpeNet-Bic, perform worse than ARHS ones, as the details generated by bicubic interpolation may not properly represent the true details, leading to suboptimal performance. Furthermore, CR^{2}F-INR2 model performs better than CR^{2}F-INR1, as its dual-branch network better leverages spatial–spectral information, enabling more comprehensive feature extraction and enhancing performance. Then, we perform a visual evaluation on those three datasets. Figs. 9–11 display the visual results graphically. As we can see, with the increasing PNN scale, the PNN results of our CR^{2}F-INR2 maintain high quality even at larger scales.

C. Full-Resolution ARHS PNN Experiments

Full-resolution experiments are used to verify the effectiveness of our proposed methods when applying ARHS PNN on the observed LRHS data and the observed PAN data. To validate the inferring abilities of our proposed CR^{2}F-INR1 and CR^{2}F-INR2 on observed LRHS/PAN (i.e., full-resolution data), full-resolution experiments are conducted on the two real datasets (i.e., the Changping dataset and the Merced dataset). Specifically, in the full-resolution inferring stage, we set $s' = 1.0, 3.0, 5.2$ as intermediate scales to make visual evaluation, wherein $s' = 1$ is also for quantitative assessment. As the resolution ratios of PAN to LRHS are 3 for Changping and Merced, there are final PNN scales $s = 3s'$ (i.e., $s = 3.0, 9.0, 15.6$) for the two datasets. Similar to the reduced-resolution experiments, the ARHS PNN is degenerated to a standard PNN if the intermediate scale $s' = 1$ in our experimental setting.

Figs. 12 and 13 give the visualization results of all 12 methods on the Changping dataset and the Merced dataset. These two figures graphically display RGB images composed of three visible bands, as well as false-color images composed of

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Fig. 12. Visualization of ARHS PNN on the real Changping dataset. (a) GSA-Bic. (b) SFIM-Bic. (c) GFPCA-Bic. (d) MTF-GLP-Bic. (e) BayesSparse-Bic. (f) PNN-Bic. (g) HyperPNN-Bic. (h) HSpeNet-Bic. (i) ARHS-CNN. (j)VSP-CNN. (k) CR^{2}F-INR1. (l) CR^{2}F-INR2.

three bands out of the visible range from pansharpened HRHS data. Obviously, our CR^{2}F-INR1 and CR^{2}F-INR2 exhibit more natural edges and finer details, with more consistent and realistic colors compared to other methods. Moreover, from these two figures, especially Fig. 12, we observe that the model-driven methods yield blurred outcomes, which implies failing to preserve spatial details during the PNN process. Though the other deep-learning-based methods, including PNN-Bic, HyperPNN-Bic, HspeNet-Bic, ARHS-CNN, and VSP-CNN, achieve better ARHS PNN results, which still suffer from color distortion. In addition to the visual comparison, we also conducted a full-resolution evaluation of all methods

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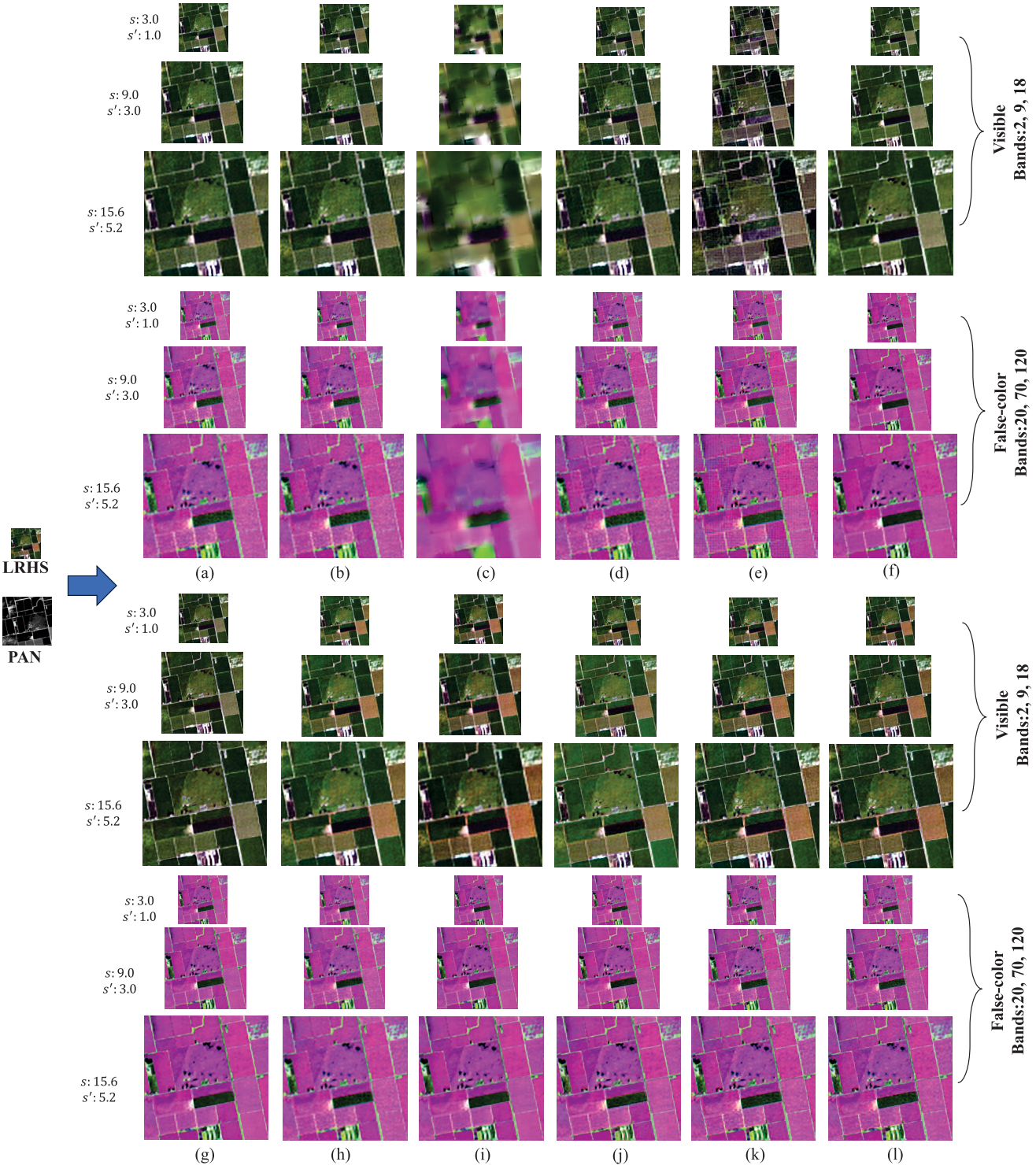


Fig. 13. Visualization of ARHS PNN on the real Merced dataset. (a) GSA-Bic. (b) SFIM-Bic. (c) GFPCA-Bic. (d) MTF-GLP-Bic. (e) BayesSparse-Bic. (f) PNN-Bic. (g) HyperPNN-Bic. (h) HSpeNet-Bic. (i) ARHS-CNN. (j) VSP-CNN. (k) CR^{2}F-INR1. (l) CR^{2}F-INR2.

using D_s, D_λ, and QRN metrics on the Merced dataset and Changping dataset. As shown in Tables II and III, our proposed methods, especially CR^{2}F-INR2, perform better than other PNN methods. From the experimental results, we can verify the excellent ARHS PNN performances of our innovatively designed ARHS PNN instances.

The above experimental results prove the effectiveness of our newly designed CR^{2}F-INR1 and CR^{2}F-INR2 for the ARHS PNN task, and meanwhile verify the superiority of our proposed CR^{2}F paradigm. Furthermore, CR^{2}F-INR2 tends to perform better than CR^{2}F-INR1 to some degree according to the experimental results, which may be due to the

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dual-path PNN INR architecture design more adapting the spectral–spatial characteristics of HS images.

VI. Conclusion and Future Lines

ARHS PNN is significantly distinct from standard HS PNN, which employs the fusion of PAN and LRHS images to produce HRHS images not only in multiple resolutions but also far beyond the PAN resolution. To deal with the task, we first propose an innovative CR^{2}F ensemble paradigm for ARHS PNN INRs; and then further develop two specific CR^{2}F ensemble-based PNN INRs, including CR^{2}F-INR1 and CR^{2}F-INR2. Both of them are basically built on ensembles with progressive structures, but CR^{2}F-INR1 relies on trunk-skip connections while CR^{2}F-INR2 is with the consideration of PAN/LRHS separately guided branches. Our newly designed ARHS PNN methods have achieved excellent PNN results on real datasets.

In our current method design, physical factors such as MTF-connected degradation are not explicitly considered. However, they indeed deserve a deep exploration for better PNN mechanism interpretation. In the future, we may enhance our work from such a perspective.

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